Aditya Rao

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Education

McMaster University Sep 2023 – Apr 2028 (Expected)
B.Eng in Software Engineering (CO-OP) *Hamilton, ON*

- **3.99/4.00 CGPA** - Dean's Honour List
- **Relevant Coursework:** Data Structures and Algorithms, Intro to Software Design, Concurrent Systems Design, Software Requirements, Large System Design, Computer Architecture, Software Testing, Databases

Experience

Software Developer Intern May 2025 – Aug 2026
IBM *Markham, ON*

- Contributed **27 open-source bug fix and feature PRs** to **OpenLiberty** and shipped **100+ internal bug fix and feature PRs** for legacy developer infrastructure, **WebSphere Application Server** and **WebSphere Liberty**.
- Developed an AI tool for Liberty build failures, automating root-cause log classification to save ∼**4 hours per failure**.
- Modernized legacy tools for **400+ developers** into Java 21 data and event-driven microservices, cutting artifact publish times from ∼**120 min to <1 min** via **Kafka messaging**.
- Engineered a work-distribution tracking platform replacing spreadsheets to save ∼**60 min/wk per manager**.

Teaching Assistant - Digital Systems Sep 2026 – Dec 2026
McMaster University *Hamilton, ON*

- Guided **50+ students** in digital logic design, RTL simulation, and synthesis using **Verilog** and **Quartus**.
- Debugged hardware-software integration on **Altera DE1-SoC FPGAs** and verified **Embedded C** binaries via Eclipse.

Full Stack Developer Intern May 2024 – Aug 2024
Medium AI *Hamilton, ON*

- Engineered real-time doctor-patient voice-based transcription with **gRPC** and **Firebase**, achieving **sub-500ms streaming latency** by load balancing **OpenAI Whisper**.
- Integrated **AI diagnostic hooks** to analyze transcripts, highlighting clinical insights for attending physicians.

Projects

BYoAI Deployment Platform | | *Go, TypeScript, Docker, MCP, OpenTelemetry*

- Built an open-source platform deploying AI agents in sandboxed **Docker** environments, using a **Go** daemon (**40 MB RSS**) for isolated execution and a **TypeScript/Bun** service for model orchestration and tool routing.
- Engineered pluggable **event routing** across Redis, RabbitMQ, and Kafka, and a custom **MCP-over-RPC transport** tunneling tool servers inside remote Docker containers.
- Built a React 19 web UI and Go terminal TUI with OAuth2 (confidential + PKCE), declarative tool ACLs, and human-in-the-loop approval gates; benchmarked security intercepts at **486ms P50**.

WebCraft | | *C++, CMake, Vcpkg, Asynchronous Programming, Github Actions*

- Authored an asynchronous runtime in **C++23** featuring asynchronous timers, task scheduling, and high-performance file and socket (TCP and UDP) I/O modules.
- Sustained **160K req/s** on synthetic **wrk** HTTP benchmarks via an **io_uring** event loop on Linux.
- Automated CI using **gtest (130+ tests)**, **GitHub Actions CI/CD** (Linux, Windows, macOS).

Technical Involvement

Sys-Admin and Fullstack Developer Executive Nov 2024 – Present
IEEE McMaster Student Branch *Hamilton, ON*

- Set up an **RPi 3B+ cluster** handling up to 100 concurrent clients with **<50ms response latency**.
- Provisioned **CI/CD pipelines** to build and deploy container images to a **Kubernetes cluster**.
- Developed the IEEE branch website using Next.js with GitHub Projects/Issues as an admin CMS.

Technical Skills

Languages: Java, JavaScript/TypeScript, Python, C#, C++, Golang, SQL, C, Verilog
Frameworks: Spring Boot, Jakarta EE, MicroProfile, Quarkus, Hibernate, Express, Next.js, FastAPI, Flask
Developer Tools: AWS, GCP, Git, Docker, Kafka, OpenShift, Kubernetes, Redis, PostgreSQL, Infinispan